Programming (C++/Java/Python)

1. Which of the following is not a primitive data type in Java?
A) int
B) float
C) String
D) char

2. Which symbol is used to terminate a statement in C++?
A) :
B) ;
C) ,
D) .

3. In Python, which function is used to check the type of a variable?
A) type()
B) typeof()
C) check()
D) var()

4. Which data type is used to store True or False values?
A) int
B) boolean
C) char
D) float

5. Which keyword is used to declare a constant variable in C++?
A) constant
B) const
C) fixed
D) define

6. Which data type has the largest size in C++?
A) int
B) float
C) double
D) char

7. In Python, variables are created when:
A) They are declared
B) They are assigned a value

C) Program starts
D) They are imported

8. Which of the following is a valid variable name?
A) 2value
B) value_2
C) value-2
D) class

9. Which keyword is used to define a class in Java?
A) struct
B) define
C) class
D) object

10. In C++, which data type is used to store a single character?
A) char
B) string
C) text
D) character

11. Which function prints output in Python?
A) echo()
B) print()
C) cout
D) display()

12. Which operator is used to access members of an object in Java?
A) :
B) ->
C) .
D) #

13. In C++, the size of char is:
A) 1 byte
B) 2 bytes
C) 4 bytes
D) 8 bytes

14. Python supports which type of typing?
A) Static
B) Dynamic

C) Strict
D) Manual

15. Which keyword is used to declare variables in Java before Java 10?
A) var
B) let
C) datatype name
D) dim

16. Which of the following is mutable in Python?
A) tuple
B) string
C) list
D) int

17. Which of the following is not a Python data type?
A) set
B) tuple
C) arraylist
D) dictionary

18. In C++, variables must be declared:
A) After use
B) Before use
C) Anywhere
D) Automatically

19. Which Java data type is used for decimal numbers?
A) int
B) float
C) boolean
D) char

20. Which Python data type is used for key-value pairs?
A) List
B) Tuple
C) Dictionary
D) Set

21. Which operator is used for addition?
A) +
B) *

C) %
D) /

22. Which operator gives remainder in C++?
A) /
B) %
C) mod
D) //

23. In Python, which operator is used for exponentiation?
A) ^
B) **
C) ^^
D) exp

24. What will be the output?
5 + 3 * 2
A) 16
B) 11
C) 13
D) 10

25. Which operator checks equality?
A) =
B) ==
C) ===
D) !=

26. Which operator is logical AND?
A) &&
B) ||
C) !
D) &

27. Which operator increases value by 1 in C++?
A) ++
B) +1
C) inc
D) add

28. The operator != means:
A) equal
B) not equal

C) greater than
D) less than

29. Which operator is used for assignment?
A) ==
B) =
C) :=
D) ->

30. Which operator is used for logical OR?
A) &&
B) ||
C) !!
D) |

31. Which operator is used for floor division in Python?
A) /
B) %
C) //
D) **

32. Which operator has highest precedence?
A) *
B) +
C) ()
D) -

33. What will be the result?
10 % 3
A) 1
B) 2
C) 3
D) 0

34. Bitwise AND operator is:
A) &
B) &&
C) |
D) ||

35. Which operator negates a boolean value?
A) ~
B) !

C) -
D) not

36. In Python logical AND is written as:
A) &&
B) and
C) &
D) &&&

37. Which operator compares greater than?
A) >
B) <
C) >=
D) <=

38. Which operator multiplies two numbers?
A) *
B) x
C) mult
D) &

39. Which operator subtracts values?
A) -
B) +
C) *
D) /

40. Which operator returns True if both conditions are true?
A) AND
B) OR
C) NOT
D) XOR

41. Which statement is used for decision making?
A) loop
B) if
C) print
D) input

42. Which loop executes at least once?
A) for
B) while

C) do-while
D) foreach

43. Which loop is used when number of iterations is known?
A) for
B) while
C) do
D) repeat

44. In Python, indentation is used to:
A) improve readability
B) define blocks
C) separate statements
D) none

45. Which keyword exits a loop?
A) stop
B) break
C) exit
D) end

46. Which keyword skips the current iteration?
A) break
B) continue
C) pass
D) stop

47. Which keyword defines a function in Python?
A) func
B) function
C) def
D) define

48. In Java, switch statement is used for:
A) looping
B) selection
C) printing
D) input

49. Which loop checks condition before execution?
A) while
B) do-while

C) both
D) none

50. What will happen in infinite loop?
A) program stops
B) program repeats forever
C) error occurs
D) compile error

51. Which keyword ends a switch case in C++?
A) stop
B) break
C) end
D) finish

52. Which statement is used to handle multiple conditions?
A) nested if
B) if-else-if
C) switch
D) all

53. Python loop over list uses:
A) for
B) iterate
C) loop
D) repeat

54. Which keyword does nothing but acts as placeholder in Python?
A) break
B) pass
C) continue
D) stop

55. A loop inside another loop is called:
A) double loop
B) nested loop
C) inside loop
D) repeated loop

56. Which loop condition must be boolean?
A) while
B) if

C) for
D) all

57. In C++ infinite loop example:
A) while(true)
B) for(;;)
C) both
D) none

58. Python else can be used with:
A) if
B) loops
C) both
D) none

59. Which structure repeats instructions?
A) sequence
B) selection
C) iteration
D) none

60. The simplest control structure is:
A) sequence
B) selection
C) loop
D) decision

61. In Java loop variable initialization is in:
A) condition
B) for statement
C) body
D) end

62. Which statement tests condition after execution?
A) while
B) do while
C) for
D) if

63. If statement belongs to:
A) selection structure
B) loop
C) function

D) variable

64. Which loop is entry-controlled?
A) for
B) while
C) both
D) do-while

65. Nested if means:
A) if inside if
B) loop inside loop
C) if inside loop
D) loop inside if

66. Python range() function is used in:
A) loops
B) conditions
C) variables
D) functions

67. Which structure allows multiple choices?
A) switch
B) for
C) while
D) do

68. Which loop works with collections in Python?
A) for
B) while
C) do
D) repeat

69. Which statement ends program immediately?
A) exit()
B) break
C) stop
D) finish

70. The loop body executes how many times in while(false)?
A) once
B) zero

C) infinite
D) two

71. Which structure is also called decision structure?
A) selection
B) iteration
C) sequence
D) none

72. Which loop is best for counting?
A) for
B) while
C) do
D) if

73. If no condition matches in switch, which block executes?
A) else
B) default
C) none
D) break

74. Logical operators are used in:
A) expressions
B) loops
C) decisions
D) all

75. Programming languages follow which control structures?
A) sequence
B) selection
C) iteration
D) all

76. What is the correct syntax to declare a function in C++?
A) function int add()
B) int add()
C) add() int
D) function add int

77. A function that does not return a value uses which keyword?
A) null
B) void

C) empty
D) return

78. Which statement is used to call a function?
A) invoke
B) run
C) function name with parentheses
D) execute

79. Which of the following allows a function to have multiple definitions with different parameters?
A) Function overriding
B) Function overloading
C) Function hiding
D) Function recursion

80. What is recursion?
A) Function calling another function
B) Function calling itself
C) Function returning multiple values
D) Function with no parameters

81. Default arguments in functions must appear:
A) Anywhere
B) At the beginning
C) At the end of parameter list
D) In the middle

82. Which object is used to display output in C++?
A) cin
B) cout
C) get
D) input

83. Which operator is used with cout?
A) >>
B) <<
C) ==
D) <>

84. Which object is used to take input from the keyboard?
A) cin
B) cout

C) input
D) scan

85. Which operator is used with cin?
A) <<
B) >>
C) ==
D) <=

86. Which header file is required for basic input/output operations?
A) stdlib.h
B) conio.h
C) iostream
D) string

87. Which manipulator is used to insert a new line?
A) endl
B) newline
C) nextline
D) break

88. Which header file is required for using strings in C++?
A) text
B) string
C) char
D) cstring

89. Which function returns the length of a string?
A) size()
B) length()
C) Both A and B
D) count()

90. Which function is used to combine two strings?
A) join()
B) append()
C) add()
D) merge()

91. Which function is used to read a full line of text?
A) cin >>
B) getline()

C) read()
D) scan()

92. Strings in C++ are stored as:
A) Numbers
B) Boolean
C) Sequence of characters
D) Floating values

93. Which operator can concatenate strings?
A) +
B) *
C) %
D) /

94. An array stores:
A) Different types of values
B) Same type of values
C) Only integers
D) Only characters

95. What is the index of the first element of an array?
A) 1
B) 0
C) -1
D) Depends on compiler

96. What is the correct declaration of an array?
A) int arr(5);
B) int arr[5];
C) array int[5];
D) arr int[5];

97. How do you access the third element of an array named arr?
A) arr(3)
B) arr[3]
C) arr[2]
D) arr[1]

98. Which loop is commonly used to traverse arrays?
A) if loop
B) while loop

C) for loop
D) switch

99. What will be the size of an array declared as int arr[10];?
A) 9
B) 10
C) 11
D) Compiler dependent

100. Which C++ container works like a dynamic array?
A) stack
B) queue
C) vector
D) set

101. In Java, functions are called:
A) Procedures
B) Methods
C) Modules
D) Classes

102. What is the correct method declaration in Java?
A) function int add()
B) int add()
C) method add()
D) add int()

103. Which keyword is used to return a value from a method?
A) send
B) give
C) return
D) result1

104. A method that does not return a value uses which keyword?
A) null
B) empty
C) void
D) static1

105. What is method overloading?
A) Same method name with different parameters
B) Same method name with same parameters

C) Different method names with same parameters
D) Calling a method twice

106. What is recursion in Java?
A) A loop inside method
B) A method calling itself
C) A method calling another method
D) A method returning multiple values

107. Which class is commonly used to take input from the user in Java?
A) Reader
B) Scanner
C) InputReader
D) ConsoleReader

108. Which package contains the Scanner class?
A) java.io
B) java.lang
C) java.util
D) java.text1

109.
A) Scanner sc = new Scanner();
B) Scanner sc = new Scanner(System.in);
C) Scanner sc = input(System.in);
D) Scanner sc = System.input();

110. Which method is used to read an integer using Scanner?
A) next()
B) nextLine()
C) nextInt()
D) getInt()

111. Which statement prints output in Java?
A) print()
B) System.print()
C) System.out.println()
D) cout<<

112. What does println() do?
A) Prints text without newline
B) Prints text with newline

C) Reads text
D) Deletes text

113. Which class is used to store strings in Java?
A) text
B) String
C) Char
D) Letters

114. Which method returns the length of a string?
A) size()
B) count()
C) length()
D) getSize()

115. Which method is used to convert a string to lowercase?
A) lower()
B) toLowerCase()
C) small()
D) lowerCase()

116. Which method is used to compare two strings?
A) compare()
B) equals()
C) match()
D) check()

117. Which method extracts a part of a string?
A) substr()
B) substring()
C) slice()
D) cut()

118. Strings in Java are:
A) Mutable
B) Immutable
C) Changeable
D) Flexible

119. Which symbol is used to declare an array in Java?
A) ()
B) []

C) {}
D) <>

120. What is the index of the first element in an array?
A) 1
B) -1
C) 0
D) 2

121. Which is the correct array declaration?
A) int arr(5);
B) int arr[5];
C) int[] arr = new int[5];
D) array int[5];

122. Which property returns the size of an array?
A) size
B) length
C) count
D) total

123. Which class represents a dynamic array in Java?
A) List
B) ArrayList
C) Set
D) Stack

124. Which package contains ArrayList?
A) java.lang
B) java.io
C) java.util
D) java.text

125. Which method adds an element to an ArrayList?
A) put()
B) add()
C) insert()
D) push()

126. Which keyword is used to define a function in Python?
A) function
B) def

C) func
D) define

127. What is the correct syntax to define a function?
A) function add():
B) def add():
C) define add():
D) add def():

128. Which keyword is used to return a value from a function?
A) send
B) give
C) return
D) output

124. What happens if a Python function does not include a return statement?
A) Error occurs
B) Returns 0
C) Returns None
D) Returns false1

125. What is recursion?
A) Function calling another function
B) Function calling itself
C) Loop inside function
D) Function returning multiple values

126. What are default parameters in Python functions?
A) Parameters with fixed values
B) Parameters with predefined values
C) Parameters that cannot change
D) Parameters without values

127. Which function is used to take input from the user in Python?
A) input()
B) read()
C) scan()
D) get()

128. Which function is used to display output?
A) show()
B) print()

C) display()
D) output()

129. What type of data does input() return by default?
A) Integer
B) Float
C) String
D) Boolean

130. How do you convert user input to an integer?
A) int(input())
B) integer(input())
C) convert(input())
D) num(input())

131. Which symbol is used to format strings in modern Python?
A) %
B) &
C) f
D) $

132. What will print("Hello", "World") output?
A) HelloWorld
B) Hello World
C) Hello,World
D) Error

133. Strings in Python are enclosed in:
A) Quotes
B) Brackets
C) Parentheses
D) Braces

134. Which function returns the length of a string?
A) count()
B) size()
C) len()
D) length()

135. Which method converts a string to lowercase?
A) lower()
B) small()

C) down()
D) lowerCase()

136. Which method converts a string to uppercase?
A) big()
B) capital()
C) upper()
D) up()

137. Which method splits a string into a list?
A) divide()
B) split()
C) break()
D) cut()

138. Which operator is used to concatenate strings?
A) +
B) *
C) %
D) /

139. Which data structure in Python is similar to an array?
A) List
B) Set
C) Tuple
D) Dictionary

140. What is the index of the first element in a list?
A) 1
B) -1
C) 0
D) 2

141. Which symbol is used to create a list?
A) ()
B) []
C) {}
D) <>

142. Which method adds an element to the end of a list?
A) add()
B) insert()

C) append()
D) push()

143. Which method removes the last element from a list?
A) remove()
B) pop()
C) delete()
D) clear()

144. Which function returns the number of elements in a list?
A) length()
B) size()
C) count()
D) len()

145. Which collection stores key-value pairs in Python?
A) List
B) Set
C) Tuple
D) Dictionary

146. Which concept of OOP allows the use of the same function name for different purposes?

A. Encapsulation
B. Polymorphism
C. Inheritance
D. Abstraction

147. What does the self keyword represent in a class?

A. It refers to the class itself
B. It refers to the current object
C. It refers to parent class
D. It refers to global variables

148. What is the process of hiding internal details and showing only functionality?

A. Polymorphism
B. Abstraction
C. Encapsulation
D. Inheritance

149. Which access specifier in C++ makes members accessible only within the same class?

A. public
B. protected
C. private
D. global

150. Which keyword is used to create an object in C++ dynamically?

A. create
B. malloc
C. new
D. object

151. Which keyword is used to inherit a class in Java?

A. implement
B. extends
C. inherits
D. derive

152. Which keyword is used to refer to the current object in Java?

A. this
B. self
C. object
D. current

153. Java supports which type of inheritance for classes?

A. Multiple inheritance
B. Single inheritance only
C. Multilevel inheritance only
D. Multiple and hierarchical inheritance

154. Method overriding is an example of:

A. Compile-time polymorphism
B. Run-time polymorphism
C. Encapsulation
D. Abstraction

155. Which keywords are used for exception handling in C++?
A. try, catch, throw
B. check, catch, throw

C. try, throw, finally
D. exception, catch

156. Which block is used to handle exceptions?

A. try
B. catch
C. throw
D. error

157. Which keyword is used to generate an exception?

A. catch
B. throw
C. try
D. exception

158. Which of the following is true about multiple catch blocks in C++?

A. Only one catch block is allowed
B. Multiple catch blocks can handle different exceptions
C. Catch blocks are optional
D. try block is optional

159. What happens if an exception is not caught in C++?

A. Program ignores it
B. Program terminates
C. Compiler fixes it
D. Program continues normally

160. Which header file is commonly used for standard exceptions in C++?

A. <error>
B. <exception>
C. <except>
D. <stdexcept>

161. Which keywords are used for exception handling in Python?

A. try, catch
B. try, except
C. check, catch
D. try, error

162. In Python which block always executes whether an exception occurs or not?

A. except
B. try
C. finally
D. error

163. Which exception occurs when dividing by zero?

A. ArithmeticError
B. ValueError
C. ZeroDivisionError
D. RuntimeError

164. What will happen if an exception is not handled in Python?

A. Program stops with an error message
B. Program continues
C. Compiler fixes it
D. Exception ignored

165. Which class is the superclass of all exceptions in Java?

A. Object
B. Throwable
C. Error
D. ExceptionHandler

166. Which type of exception must be handled at compile time in Java?

A. Runtime exception
B. Checked exception
C. Logical exception
D. System exception

167. In Python, a module is usually stored as:

A. .class file
B. .exe file
C. .py file
D. .obj file

168. In Python which statement imports only a specific function from a module?

A. import module
B. include module
C. from module import function
D. import function module

169. Which of the following is a built-in Python module?

A. math
B. random
C. sys
D. All of the above

170.In Python what does dir() function do when used with a module?

A. Deletes module
B. Lists module attributes
C. Compiles module
D. Runs module

171. In Python, a directory is recognized as a package if it contains:

A. main.py
B. __init__.py
C. package.py
D. module.py

172. Which syntax imports a module from a package?

A. import package.module
B. include package.module
C. module.package import
D. using package

173. Which of the following represents hierarchical organization of modules?

A. Package
B. Class
C. Function
D. Variable

174. Which of the following is a standard Java package?

A. java.util
B. java.lang

C. java.io
D. All of the above

175. Which header file contains functions for character handling?

A. ctype.h
B. char.h
C. string.h
D. text.h

176. Which header file provides dynamic memory allocation functions?

A. stdlib.h
B. memory.h
C. malloc.h
D. All of the above

177. Which type of memory allocation happens during program execution?

A. Static memory allocation
B. Dynamic memory allocation
C. Compile-time allocation
D. Fixed allocation

178. Which operator is used for dynamic memory allocation in C++?

A. malloc
B. alloc
C. new
D. create

179. Which memory area stores local variables and function calls?

A. Heap
B. Stack
C. Cache
D. Register

180. Which memory area is used for dynamic memory allocation?

A. Heap
B. Stack
C. ROM
D. Cache

181. What is a memory leak?

A. Memory used by CPU
B. Memory that is allocated but never released
C. Memory used by operating system
D. Memory stored in registers

182. Which function is used for dynamic memory allocation in C language?

A. new()
B. malloc()
C. alloc()
D. reserve()

183. What is garbage collection?

A. Deleting unused files
B. Automatic memory deallocation
C. Memory compression
D. Memory expansion

184. Which programming language automatically manages memory using garbage collection?

A. C++
B. Assembly
C. Java
D. C

185. What happens if dynamically allocated memory is not released?

A. Program runs faster
B. Memory leak occurs
C. Program stops compiling
D. Memory doubles automatically

186. Which keyword is used in Java to request garbage collection?

A. gc()
B. collect()
C. System.gc()
D. free()

187. Which memory allocation is faster?

A. Heap allocation
B. Stack allocation
C. Dynamic allocation
D. Virtual allocation

188. Which of the following problems occurs due to improper memory management?

A. Memory leak
B. Fragmentation
C. Program crash
D. All of the above

189. Fragmentation occurs when:

A. Memory is completely full
B. Memory is divided into small unused blocks
C. CPU is overloaded
D. Disk space is low

190. Which function is used to free dynamically allocated memory in C?

A. free()
B. delete()
C. release()
D. clear()

191. In OOP, objects are typically stored in which memory area?

A. Stack
B. Heap
C. Cache
D. Register

192. In C++, objects created using the new operator are stored in:

A. Stack
B. Heap
C. ROM
D. Register

193. What is a destructor in OOP?

A. Function used to create objects
B. Function used to destroy objects and release memory

C. Function used to allocate memory
D. Function used to copy objects

194. When does a destructor get automatically called?

A. When object is created
B. When object goes out of scope
C. When program starts
D. When constructor is called

195. What is a memory leak in object-oriented programs?

A. Memory accessed by multiple objects
B. Memory allocated but not released after object use
C. Memory shared by classes
D. Memory used by CPU

196. What is a dangling pointer in C++?

A. Pointer pointing to a valid object
B. Pointer pointing to deleted memory
C. Pointer pointing to a class
D. Pointer pointing to a function

197. What is method overriding?

A. Defining multiple methods with same name but different parameters
B. Redefining a parent class method in a child class
C. Hiding data members
D. Creating private methods

198. Which keyword prevents inheritance of a class in Java?

A. static
B. final
C. constant
D. private

199. Which keyword in C++ allows a function to be overridden in derived classes?

A. override
B. virtual
C. dynamic
D. static

200. What is composition in OOP?

A. A class contains objects of another class
B. A class inherits another class
C. A class hides another class
D. A class deletes another class

201. Which concept allows objects of different classes to be treated as objects of a common superclass?

A. Encapsulation
B. Polymorphism
C. Abstraction
D. Aggregation

202. What is dynamic binding?

A. Binding method calls at compile time
B. Binding method calls at runtime
C. Binding variables at compile time
D. Binding classes at compile time

203. Which inheritance type is NOT supported directly for classes in Java?

A. Multiple inheritance
B. Single inheritance
C. Multilevel inheritance
D. Hierarchical inheritance

204. Which of the following best describes aggregation?

A. Strong ownership relationship
B. Weak relationship between objects
C. Object duplication
D. Class hiding

205. Which concept improves code reusability in OOP the most?

A. Inheritance
B. Encapsulation
C. Abstraction
D. Polymorphism

206. Which OOP principle allows restricting direct access to object data?

A. Polymorphism
B. Encapsulation
C. Inheritance
D. Abstraction

207. Which type of inheritance involves one base class and multiple derived classes?

A. Multiple inheritance
B. Multilevel inheritance
C. Hierarchical inheritance
D. Hybrid inheritance

208. Which Java block is executed only if no exception occurs?

A. finally
B. else
C. try
D. catch

209. What is the main purpose of modules?

A. Improve graphics
B. Organize and reuse code
C. Increase CPU speed
D. Store memory

210. Which OOP concept allows methods with the same name but different parameters?

A. Method overriding
B. Method overloading
C. Encapsulation
D. Abstraction

211. Which concept represents a “has-a” relationship between classes?

A. Inheritance
B. Composition
C. Polymorphism
D. Abstraction

212. What is an abstract method?

A. A method with implementation
B. A method without body

C. A private method
D. A static method

213. Which block in exception handling is optional but used to execute cleanup code?

A. try
B. catch
C. finally
D. throw

214. In C++, which mechanism allows handling exceptions of different types using inheritance?

A. Virtual functions
B. Exception hierarchy
C. Templates
D. Namespaces

215. Which Python keyword is used to define a custom exception?

A. create
B. raise
C. class
D. error

216. What is the purpose of namespaces in C++?

A. Memory allocation
B. Avoid name conflicts
C. Exception handling
D. File management

217. Which memory issue occurs when recursive function calls exceed limit?

A. Heap overflow
B. Stack overflow
C. Memory leak
D. Buffer underflow

218. What is the main advantage of using smart pointers in C++?

A. Faster execution
B. Automatic memory management
C. Better syntax
D. Increased heap size

219. Which concept allows grouping related functions and variables under one unit?

A. Object
B. Module
C. Package
D. Library

220. In Java, interfaces support which feature?

A. Multiple inheritance of classes
B. Multiple inheritance of types
C. Dynamic memory allocation
D. Compilation control

221. Which of the following is a runtime error?

A. Syntax error
B. Logical error
C. Null pointer exception
D. Missing semicolon

222. In Java, what happens if a class implements an interface but does not implement all methods?

A. It compiles normally
B. It becomes abstract
C. It becomes final
D. It causes runtime error

223. Which programming paradigm does Java primarily support?
A) Procedural
B) Object-Oriented
C) Functional
D) Logical

224. Which class is automatically inherited by all Java classes?
A) Object
B) Base
C) System
D) Parent

225. In C++, which keyword is used to prevent modification of a variable?
A) fixed
B) const

C) final
D) static

226. Which design pattern ensures only one instance of a class exists?
A) Factory
B) Singleton
C) Builder
D) Adapter

227. Which term refers to executing multiple tasks simultaneously?
A) Compilation
B) Parallelism
C) Interpretation
D) Execution

228. In Java, threads are created using:
A) Thread class
B) Runnable interface
C) Both A and B
D) Process class

229. Which keyword is used to synchronize threads in Java?
A) lock
B) synchronized
C) thread-safe
D) atomic

230. Which module provides threading support in Python?
A) thread
B) threading
C) parallel
D) multiprocessing

231. Which problem occurs when two threads access shared data simultaneously?
A) Deadlock
B) Race condition
C) Memory leak
D) Overflow

232. Which situation occurs when two threads wait indefinitely for each other?
A) Race condition
B) Deadlock

C) Parallelism
D) Looping

233. Which C++ library supports multithreading?
A) <thread>
B) <parallel>
C) <multi>
D) <process>

234. Which Python module allows true parallel processing using multiple CPU cores?
A) threading
B) multiprocessing
C) concurrent
D) async

235. Which function starts a thread in Java?
A) run()
B) start()
C) execute()
D) init()

236. Which Python keyword is used in asynchronous programming?
A) async
B) await
C) both
D) none

237. Which concept improves performance using multiple processors?
A) Multitasking
B) Parallel computing
C) Compilation
D) Virtualization

238. In Java, which package provides concurrency utilities?
A) java.thread
B) java.concurrent
C) java.util.concurrent
D) java.parallel

239. Which term refers to executing tasks in overlapping time periods?
A) Parallelism
B) Concurrency

C) Compilation
D) Optimization

240. Which function waits for a thread to finish in Java?
A) wait()
B) sleep()
C) join()
D) stop()

241. Which issue occurs when threads compete for shared resources?
A) Deadlock
B) Race condition
C) Starvation
D) All

242. Which Python library supports asynchronous programming?
A) asyncio
B) threading
C) multiprocessing
D) parallel

243. Which term refers to dividing tasks among multiple threads?
A) Scheduling
B) Parallelization
C) Compilation
D) Linking

244. Which mechanism prevents race conditions?
A) Locks
B) Mutex
C) Synchronization
D) All

245. Which Java class pauses thread execution?
A) Thread.sleep()
B) wait()
C) stop()
D) halt()

246. Which concept improves responsiveness in applications?
A) Multithreading
B) Compilation

C) Linking
D) Parsing

247. Debugging is the process of:
A) writing code
B) finding and fixing errors
C) compiling programs
D) executing programs

248. Which type of error occurs during compilation?
A) Syntax error
B) Logical error
C) Runtime error
D) Semantic error

249. Which tool helps detect runtime errors in Python?
A) debugger
B) profiler
C) interpreter
D) optimizer

250. Which type of testing checks individual program units?
A) System testing
B) Integration testing
C) Unit testing
D) Acceptance testing

251. In Python, which module is commonly used for unit testing?
A) pytest
B) unittest
C) both
D) none

252. Which type of error produces incorrect output but no crash?
A) Syntax error
B) Logical error
C) Runtime error
D) Compile error

253. Optimization aims to:
A) reduce program errors
B) improve performance

C) increase code length
D) slow execution

254. Which technique improves memory usage?
A) code duplication
B) memory optimization
C) recursion
D) compilation

255. Which debugging technique prints variable values during execution?
A) step debugging
B) logging
C) tracing
D) print debugging

256. Which testing ensures system components work together?
A) Unit testing
B) Integration testing
C) Regression testing
D) System testing

257. Which testing verifies bug fixes do not introduce new errors?
A) Regression testing
B) Unit testing
C) Integration testing
D) Functional testing

258. Which Python tool measures program performance?
A) profiler
B) debugger
C) compiler
D) optimizer

259. Code optimization focuses on:
A) faster execution
B) reduced memory usage
C) improved efficiency
D) all

260. Which testing is performed by developers?
A) Unit testing
B) Acceptance testing

C) Beta testing
D) User testing

261. A breakpoint is used in:
A) debugging
B) compiling
C) linking
D) execution

262. Which type of testing checks the entire system?
A) Unit
B) System
C) Integration
D) Regression

263. Which tool automatically runs tests after code changes?
A) CI/CD
B) Compiler
C) Debugger
D) Editor

264. Which optimization reduces repeated calculations?
A) caching
B) recursion
C) looping
D) duplication

265. Which type of testing is performed before release?
A) Alpha testing
B) Beta testing
C) Both
D) None

266. Which process improves code readability and maintainability?
A) Refactoring
B) Debugging
C) Testing
D) Compilation

267. Which methodology emphasizes iterative development?
A) Waterfall
B) Agile

C) Spiral
D) Prototype

268. Which tool tracks code changes?
A) Git
B) Compiler
C) Debugger
D) Interpreter

269. Which platform hosts Git repositories online?
A) GitHub
B) StackOverflow
C) Eclipse
D) Jenkins

270. Continuous Integration means:
A) merging code frequently
B) compiling manually
C) coding individually
D) testing rarely

271. Which practice improves code quality through review?
A) Code review
B) Compilation
C) Testing
D) Debugging

272. Documentation helps in:
A) understanding code
B) maintenance
C) collaboration
D) all

273. Which principle avoids code duplication?
A) DRY
B) KISS
C) SOLID
D) OOP

274. Which principle means “Keep It Simple”?
A) DRY
B) KISS

C) SOLID
D) YAGNI

275. Which practice ensures small, frequent updates?
A) Agile
B) Waterfall
C) Spiral
D) Prototype

276. Which document describes system functionality?
A) Requirements specification
B) Source code
C) Test case
D) Output file

277. Which tool automates building and testing software?
A) Jenkins
B) Git
C) Python
D) Java

278. Which testing verifies user requirements?
A) Acceptance testing
B) Unit testing
C) Integration testing
D) Regression testing

279. Which development model follows sequential phases?
A) Agile
B) Waterfall
C) Scrum
D) DevOps

280. DevOps combines:
A) Development and Operations
B) Design and Testing
C) Coding and Debugging
D) Analysis and Implementation

281. Code maintainability means:
A) code runs fast
B) code is easy to modify

C) code is large
D) code is complex

282.What is the primary goal of code refactoring?

A. Adding new features
B. Improving code structure without changing behavior
C. Increasing program speed
D. Fixing runtime errors

283. Pair programming is commonly used in which development methodology?

A. Waterfall
B. Spiral
C. Agile / Extreme Programming (XP)
D. V-Model

284. Which concept enables runtime polymorphism in C++?

A. Templates
B. Virtual functions
C. Namespaces
D. Constructors

285. What is the main purpose of generics/templates?

A. Improve execution speed
B. Enable type-safe reusable code
C. Manage memory
D. Handle exceptions

286. In C++, which keyword helps in debugging by checking conditions?

A. try
B. assert
C. debug
D. check

287. What does a stack trace show?

A. Memory usage
B. Compilation steps
C. Sequence of function calls leading to error
D. Variable declaration

288. Which is an example of debugging technique?

A. Code encryption
B. Print statements
C. Hardware upgrade
D. File compression

289. Debug mode in an IDE allows:

A. Faster execution
B. Step-by-step execution
C. Code encryption
D. Memory deletion

290. In C++, trace the output

```
int x = 3;

cout << x++ << " " << x;
```

A. 3 3
B. 3 4
C. 4 4
D. 4 3

291. In Python trace the output:

```
for i in range(2):

    print(i, end=" ")
```

A. 1 2
B. 0 1
C. 0 1 2
D. 2 1

292. In Java trace the output:

```
String s = "AI";

s.concat("ML");

System.out.println(s);
```

A. AIML
B. ML

C. AI
D. Error

293. In C++ trace the output:

```
int a = 7;

int *p = &a;

*p = 20;

cout << a;
```

A. 7
B. 20
C. Address
D. Error

294. In C++ trace the output:

```
int f(int n){

   return f(n-1);

}
```

A. 0
B. Error at compile time
C. Stack overflow
D. 1

295. In C++ Which operator is used to get the address of a variable?

A. *

B. &

C. ->

D. ::

296. What does a pointer store in C++?

A. Value of a variable
B. Address of a variable
C. Type of a variable
D. Reference count

297. Which of the following is TRUE about references?

A. They can be null
B. They can be reseated
C. They must refer to an existing variable
D. They consume no memory

298. In C++ trace the output:

```
int* p = new int(5);

delete p;

cout << *p;
```

A. Correct

B. Memory leak

C. Dangling pointer

D. Compilation error

299. In Python What does the yield keyword do?

A. Stops program
B. Returns value and preserves state
C. Deletes variable
D. Creates thread

300. What is an iterator in Python?

A. Object with iter and next methods
B. Loop variable
C. Function
D. List only